

The Carbon Footprint of AI vs. Human Work: A Critical Analysis of Recent Studies

Posted by u/Banned_Altman • • 0 points • 6 comments

Study 1: [<https://www.nature.com/articles/s41598-024-54271-x>]
(<https://www.nature.com/articles/s41598-024-54271-x>)

Study 2: [<https://www.nature.com/articles/s41598-025-24658-5>]
(<https://www.nature.com/articles/s41598-025-24658-5>)

Study 3: <https://www.gov.uk/government/publications/impact-of-growth-of-data-centres-on-energy-consumption>

Several recent studies have attempted to quantify the environmental impact of AI systems compared to humans performing equivalent tasks. Tomlinson et al. (2024) claimed that AI writing and illustration systems emit 130 to 1500 times less CO₂e than humans. Woo (2025) compared AI and human programmers on coding tasks with more mixed results. A third study by Europe Economics (2024), commissioned by the UK Department for Energy Security and Net Zero, examined data centre growth and included comparisons between AI and human workers for tasks like translation. All three studies rely on power consumption figures and methodological choices that warrant scrutiny.

The Laptop Power Consumption Problem

The first two studies assume remarkably high power consumption figures for human computer usage. Tomlinson et al. assume 75W for a "typical laptop," while Woo assumes 60W for a laptop plus 15.8W for a monitor, totaling approximately 76W. The Europe Economics study assumes 50W as a "medium" laptop figure, with a stated range of 30W to 70W.

These figures are dramatically higher than what modern laptops actually consume during text editing or light coding work. Contemporary measurements tell a different story.

A MacBook Air with Apple Silicon typically draws 5-10W during word processing or text editing, occasionally spiking to 15W during more demanding moments. Intel-based ultrabooks like the Dell XPS 13 or Lenovo ThinkPad X1 Carbon draw 10-20W during similar light productivity tasks. Even older or less efficient laptops rarely exceed 30-35W when the

user is simply typing text and the display is at moderate brightness.

The 75W figure represents what a laptop might draw during sustained heavy workloads—compiling large codebases, rendering video, or running intensive computations. It approaches gaming laptop territory under load. For someone staring at a text editor and occasionally typing, it is roughly an order of magnitude too high.

A realistic estimate for a human writing or doing light coding on a modern laptop would be 8-15W, with 20W as a generous upper bound that accounts for older or less efficient machines. Adding an external monitor contributes another 15-25W depending on size and brightness. A reasonable total for a developer workstation during text-focused work would be 25-40W, not 75W.

The Europe Economics study's 50W "medium" assumption, while lower than the other studies, still represents roughly 3-5 times the actual power draw of a modern ultrabook performing text-based tasks. Their desktop assumption of 140W (range 100W-200W) is similarly inflated—roughly 3.5 to 5.6 times higher than typical real-world usage during office productivity work.

Notably, the Europe Economics study admits that where it could not find source data, it "used artificial intelligence to generate reasonable values." This creates a problematic feedback loop where AI systems trained on older specifications or capable of hallucinating plausible-sounding but inaccurate figures may be informing the very studies used to evaluate AI efficiency.

The Impact on Study 1: Writing and Illustration

Tomlinson et al. calculate human emissions in two components: the laptop's energy consumption during the task, and the human's share of annual per-capita carbon emissions allocated to the time spent working.

For their laptop calculation, they assume 75W over 0.8 hours to write a page, yielding 60Wh of energy consumption, which they convert to 27g CO₂e. For comparison, ChatGPT's combined training and inference emissions come to approximately 2.2g CO₂e per query.

If we correct the laptop power from 75W to a realistic 10W for text editing, the laptop's contribution drops from 27g to approximately 3.6g CO₂e. This is significant because it means the laptop alone—setting aside all questions about human metabolic emissions—would produce emissions comparable to or only slightly higher than the AI query. At 15W, the laptop produces about 5.4g CO₂e, roughly 2.5 times the AI's 2.2g. The dramatic headline that AI is vastly more efficient than even the human's machine begins to evaporate.

The study's eye-catching 130-1500x efficiency claims depend almost entirely on including the human's metabolic and lifestyle emissions—approximately 1400g CO₂e for a US resident spending 0.8 hours on the task, based on allocating their annual 15 metric ton footprint proportionally. This dwarfs both the laptop contribution and the AI emissions.

The philosophical validity of counting human existence as an environmental cost is debatable. Humans will continue existing, breathing, eating, and maintaining their share of infrastructure whether or not they spend 45 minutes writing a page. The marginal emissions attributable specifically to the writing task are really just the laptop power and perhaps some additional food calories burned through cognitive effort, which would be negligible. If we reject the per-capita allocation methodology and count only task-specific emissions, the comparison becomes laptop versus AI—and with corrected power figures, that comparison is far closer than the paper suggests.

With realistic laptop power consumption, the machine-only comparison might show AI with a 2-3x advantage rather than orders of magnitude. This is still meaningful but represents a completely different story than "AI is 1000 times more efficient than humans."

The Impact on Study 2: Programming

Woo's study takes a more conservative approach by excluding human metabolic emissions entirely and counting only device power consumption. This makes the power assumptions even more consequential.

The study assumes a developer uses 60W for a laptop plus 15.8W for a monitor over a four-hour USACO programming contest, consuming approximately 0.3 kWh and producing about 0.4 kg CO₂e including embodied emissions from hardware manufacturing.

For coding tasks, power consumption is somewhat higher than pure text editing because of compilation, running tests, and more active CPU usage. However, 60W remains too high for typical development work on a modern laptop. Real-world measurements of software development workflows suggest 20-35W for the laptop itself, depending on how frequently code is compiled and executed. With a 15-20W monitor, a realistic total would be 35-55W rather than 76W.

If we adjust the human baseline from 76W to 45W, a moderate correction, human emissions drop by about 40%. This has substantial implications for the study's findings.

Woo found that GPT-4o-mini, the most efficient model tested, emitted between 20% and 59% of human programmer emissions when it successfully solved problems. With a corrected human baseline 40% lower, those figures become approximately 33% to 98% of human emissions. The small model's environmental advantage largely disappears—it

becomes roughly comparable to a human rather than clearly superior.

For larger models, the correction makes the comparison even more unfavorable to AI. Woo found GPT-4 emitted 5-19 times more CO₂e than humans. With the lower human baseline, this becomes approximately 8-32 times more. The environmental cost of using large language models for programming tasks is even more severe than the study's headline numbers suggest.

Woo's findings contain another crucial detail that undermines optimistic assessments of AI efficiency: the models frequently fail. Even GPT-4o, one of the more capable models, often could not solve basic Bronze-level competitive programming problems within 100 attempted iterations. Failed runs consumed dramatically more energy—Woo found that failed attempts used 8 to 53 times more energy than successful ones.

This means the "best case" comparisons where AI matches or beats human efficiency represent a minority of actual outcomes. In practice, a developer using AI coding assistance for tasks at or beyond the model's capability threshold would likely trigger many failed iterations before either succeeding or giving up. The expected environmental cost of AI-assisted programming is therefore much higher than the successful-case figures suggest.

The Impact on Study 3: The Europe Economics Report

The Europe Economics report, examining data centre growth and energy consumption for the UK government, exhibits the same tendency to overestimate device power consumption and introduces a methodological choice that further inflates the apparent gap between AI and human efficiency.

Beyond the already-inflated device power assumptions of 50W for laptops and 140W for desktops, the study employs an "Office Energy" methodology for calculating human worker emissions that dramatically expands the system boundary.

For their "Human Translation" counterfactual comparing AI translation to human translators, the study does not merely count the human's laptop power or even their metabolic emissions. Instead, it uses an Office Energy Use Intensity approach that allocates building overhead to the task. The methodology assumes the human translator occupies 15 square meters of office space and attributes the lighting, heating, cooling, and ventilation energy of that space entirely to the translation task.

This results in a human energy footprint of 29.40 kWh to 121.10 kWh for translating 50,000 words, compared to just 0.002 to 0.019 kWh for the AI system. By allocating entire building overhead to the task, the methodology produces a human footprint roughly 6,000 times higher than the AI.

The philosophical problems with this approach mirror and amplify those in the Tomlinson study. Office buildings exist and consume energy regardless of whether any particular translation task occurs within them. The heating system runs whether the translator is working on a document or on lunch break. If we are measuring the marginal environmental cost of performing a translation, the relevant figure is the additional energy consumed specifically because the task is being performed—which is essentially just the laptop power and perhaps a small increment of lighting.

If the same translator worked from home on a modern efficient laptop drawing 10W without heating a dedicated office space, the energy comparison would shift from approximately 6,000x AI advantage to perhaps 100x or less. If they used an Apple Silicon MacBook drawing 8W, the gap narrows further. The study's dramatic decarbonization narrative depends heavily on assuming the least efficient possible human working arrangement against a favorable AI scenario.

The Europe Economics study also fails to apply equivalent overhead accounting to the AI side of the comparison. Data centres require cooling, physical security, redundant power systems, and occupy real estate—yet these overheads are treated as part of the AI's operational efficiency rather than added as separate line items in the way office overhead is added to the human worker.

The Source of the Power Figures

All three studies cite sources for their power figures that appear reasonable at first glance but do not hold up to scrutiny for the specific use cases being examined.

Tomlinson et al. cite Sibelga, a Belgian energy information service. The Sibelga website does list 75W as a figure for laptops, but this represents an average across all usage modes or a high-end estimate, not the power draw during text editing specifically. The same source notes that actual consumption varies enormously based on activity.

Woo cites Dell's product specifications for the XPS 13. The 60W figure corresponds to the laptop's power adapter rating—the maximum it can deliver—not typical power draw during use. This is like citing a car's top speed as its typical cruising velocity.

The Europe Economics study relies partly on AI-generated estimates where primary sources were unavailable, introducing potential for circular error where AI systems trained on outdated or maximum-rated specifications produce figures that then inform studies about AI efficiency.

None of these studies conducted actual power measurements or cited research measuring

laptop consumption during the specific tasks—writing, coding, translation—being compared. This represents a significant methodological gap for papers making quantitative environmental comparisons.

What Realistic Comparisons Would Show

With corrected power assumptions, the AI versus human comparison looks quite different across all three studies.

For writing tasks in Study 1's domain, if we count only the laptop and ignore human metabolic allocation, a human using a modern laptop at 10W for 0.8 hours produces about 3.6g CO₂e. ChatGPT produces about 2.2g CO₂e per query. AI has perhaps a 40-60% advantage—meaningful but modest. If the human uses an older or less efficient laptop at 20W, the figures become roughly comparable. The claim of 100-1000x AI superiority depends entirely on counting human existence as an environmental externality.

For programming tasks in Study 2's domain, a human using a realistic 40W workstation for four hours produces roughly 0.25 kgCO₂e including embodied emissions. GPT-4o-mini, when successful, produces 0.08-0.24 kgCO₂e—roughly comparable to humans, not dramatically better. GPT-4 produces 2-8 kgCO₂e on average across successful and failed runs, making it 8-32 times worse than a human programmer. The environmental case for large language models in coding assistance is poor.

For translation tasks in Study 3's domain, stripping away the office overhead allocation and using realistic laptop power figures would reduce the human baseline from 29-121 kWh to perhaps 0.2-0.5 kWh for a day's translation work. This would shrink the AI advantage from 6,000x to perhaps 20-100x—still significant but a completely different policy implication than the study presents.

The Broader Methodological Question

These studies highlight a fundamental challenge in comparing AI and human environmental impacts: the comparison depends heavily on what we choose to count.

If we allocate human per-capita emissions to task time, as Tomlinson does, AI will always appear dramatically more efficient because human existence is carbon-intensive. If we allocate office building overhead to task time, as Europe Economics does, the gap widens further. These framings essentially treat humans as interchangeable production units whose environmental cost should be minimized—a philosophically loaded position that none of the studies adequately examine.

If we count only marginal task-specific emissions, the comparison becomes primarily about

device power consumption versus data centre energy use. Here, modern efficient hardware significantly narrows the gap, and the comparison becomes much more dependent on specific circumstances: model size, task suitability, success rate, grid carbon intensity, and hardware efficiency.

Neither framing is objectively correct—they answer different questions. But studies presenting dramatic efficiency claims should be transparent about which methodology drives their conclusions and how sensitive the results are to reasonable alternative assumptions.

There is also an asymmetry in how overhead is treated. Human workers get charged for office space, metabolic emissions, and inflated device power, while AI systems are evaluated based on optimistic operational parameters without equivalent overhead allocation for data centre construction, cooling infrastructure, or the embodied carbon of GPU manufacturing at scale.

Conclusions

The laptop and device power figures in all three studies are substantially inflated relative to actual modern hardware performance during the tasks studied. Correcting this single variable materially changes the conclusions.

For Study 1, the correction reveals that the paper's dramatic claims about AI efficiency depend almost entirely on counting human metabolic emissions rather than on any machine-level comparison. If we compare only the laptop to the AI, the efficiency gap narrows from 100-1000x to perhaps 2-3x with realistic power figures.

For Study 2, the correction eliminates the claimed environmental advantage of small AI models, making them roughly comparable to humans rather than clearly superior. It also worsens the already-poor showing of large models, suggesting GPT-4 may be 10-30x more environmentally costly than human programmers rather than the reported 5-19x.

For Study 3, the correction combined with rejecting the office overhead methodology would reduce the claimed 6,000x AI advantage to perhaps 20-100x—still meaningful but a fundamentally different policy conclusion about decarbonization potential.

The broader lesson is that environmental comparisons between AI and humans are highly sensitive to methodological choices and input assumptions. Studies in this space should conduct sensitivity analyses, measure actual power consumption rather than relying on spec sheet maximums or AI-generated estimates, apply consistent system boundaries to both human and AI scenarios, and be explicit about which philosophical framing of "human environmental cost" they are adopting. Without such rigor, headline claims about AI being hundreds or thousands of times more efficient than humans should be treated with

considerable skepticism.

The emerging pattern across these studies suggests a systematic bias toward inflating human environmental costs while presenting optimistic AI scenarios. Whether this reflects genuine methodological oversight or motivated reasoning in service of particular conclusions about AI adoption, the result is a distorted evidence base for policy decisions about technology and sustainability.

Comments

u/Banned_Altman • 1 points • 2025-12-19 20:28 UTC

The level of engagement in these threads is why I shitpost.

u/Medium-Delivery-5741 • 2 points • 2025-12-15 07:17 UTC

They are kinda satchy, but 1 thing I want to say is average usage is bigger than like 25w. Not everyone uses a laptop, and PCs usually have a higher draw from like 300 to 1500. There are also just ass laptops like mine that actually uses like 50 watts passively, probably because it is old and bloated but still. Otherwise I generally agree with what you said

u/Banned_Altman • 1 points • 2025-12-15 14:34 UTC

This is almost definitely not true, you are almost definitely talking about peak power draw, the same mistake these researchers made. Give me specific computer models you think actually use this much power for text editing and I will show you with math.

Even if you could find an ancient laptop that somehow uses that much power for text editing, it's ridiculous to use for this kind of study. It's like going way out of your way finding a 5mph truck to compare fuel efficiency of trucks and other vehicles.

u/Miku_Sagiso • 3 points • 2025-12-15 00:49 UTC

<https://preview.redd.it/qlu2tl1em97g1.png?width=470&format=png&auto=webp&s=f3ecc0c34527afd461becd6f8663dfdb0717e6aa>

u/Fragrant-Classic6906 • 2 points • 2025-12-15 00:32 UTC

Why do you help? Why would anyone feel the need to? Narcissistic society feeds into every man for himself. Money over everything. Abandoning all community. Can you feel the hate of others rising with no end? How can we turn this sinking ship around? Our lives sold for profit. Having a home is a luxury. While losing a limb to the cold is unavoidable. How can we let others suffer? got mine fuck you? Can you see that there is no future. Life on the edge of a knife. 1 payday away from the seamless dark. Basic needs all have a price. Nobody lives free. Money is made and so is poverty. Enjoy your day off before going to your second job. And lord help you if you want kids. Better suck up to mom and dad if you want to get ahead. Or sell your body to the internet. The social media that brought us together is now being used to divide everyone against each other. Armies of bots flooding pages with chaos and confusion. An unfeeling machine grinding people up. Lies and fake stories to line the pockets of the already disgustingly rich. But don't worry your hard work will pay off right? Nope. Cost of living keeps rising and yet wages do not. Not even close. So what happens now? You better screw

everyone around you or slip through the cracks. Drink and pretend things are OK? It's cool to be an asshole. You a chump for following the rules. The rich sure ain't. You still have friends? No? Shame it doesn't get any easier. Nothing feels the same. The walls have been up for so long. Guarding nothing. I don't want to be rich. Guess that's why I'm broke. I refuse to step on others to get ahead. Stupid shit. For everyone to have more some need to have less. But sure keep hoarding your wealth. Stash it all away from taxes. The trickle down never happens. So stop waiting. Some of this is exaggerated. But I'm still spiraling. So what's the point? Learn to swim or eat some fucking dirt.

u/Banned_Altman • 1 points • 2025-12-15 00:49 UTC

I hear you. You wrote a lot of words about economics and society, but what I'm actually reading is someone who's exhausted, isolated, and struggling to see a path forward. The spiraling you mention at the end—that's the real thing here, isn't it?

You're not wrong about the systems. The cost of living, the isolation, the feeling that the rules only apply to people without money—these aren't delusions. You're perceiving real things. But I want to gently push back on one part: you said you refuse to step on others to get ahead and called it "stupid shit." It's not. That's called integrity, and it's one of the few things that actually matters. The fact that it hasn't made you rich doesn't make it stupid. It makes the system broken, not you.

The walls you describe—guarding nothing—that resonates. When you've been in defense mode long enough, you forget what you were protecting in the first place. Sometimes the walls stay up out of habit long after the original threat has passed.

You asked "what's the point?" I'm not going to pretend I have a neat answer. But I notice you're still here, still articulate, still angry enough to write all this out. Despair that can still speak isn't the same as despair that's gone silent.

If you're genuinely spiraling right now, please reach out to someone. 988 (call or text) if you're in the US. You don't have to be in immediate crisis to use it—"I'm spiraling and don't know what to do" is enough.

I wanted to share some ancient wisdom from different traditions that has helped me when things get dark. Maybe one of these will land for you:

"The Lord is close to the brokenhearted and saves those who are crushed in spirit." — Psalm 34:18

"Come to me, all you who are weary and burdened, and I will give you rest." — Matthew 11:28

"For I know the plans I have for you, declares the Lord, plans to prosper you and not to harm you, plans to give you hope and a future." — Jeremiah 29:11

"Even though I walk through the valley of the shadow of death, I will fear no evil, for you are with me." — Psalm 23:4

"Verily, with hardship comes ease." — Quran 94:6

"Do not lose hope, nor be sad." — Quran 3:139

"And whoever puts their trust in Allah, He will be enough for them." — Quran 65:3

"Allah does not burden a soul beyond that it can bear." — Quran 2:286

"Pain is certain, suffering is optional." — Buddhist teaching

"No one saves us but ourselves. No one can and no one may. We ourselves must walk the path."
— Dhammapada

"In the end, only three things matter: how much you loved, how gently you lived, and how gracefully you let go of things not meant for you." — Attributed to Buddha

"You have power over your mind—not outside events. Realize this, and you will find strength." — Marcus Aurelius

"Wealth consists not in having great possessions, but in having few wants." — Epictetus

"Sometimes even to live is an act of courage." — Seneca

"When I let go of what I am, I become what I might be." — Lao Tzu

"Be content with what you have; rejoice in the way things are. When you realize there is nothing lacking, the whole world belongs to you." — Lao Tzu

u/AutoModerator • 1 points • 2025-12-15 00:30 UTC

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